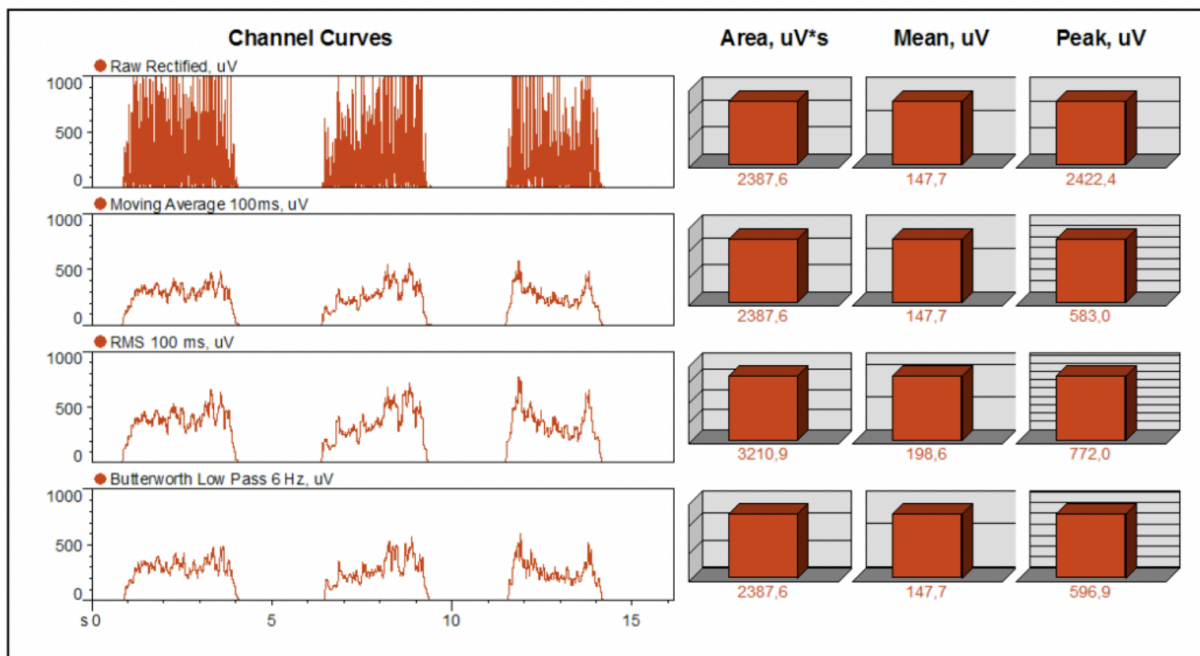


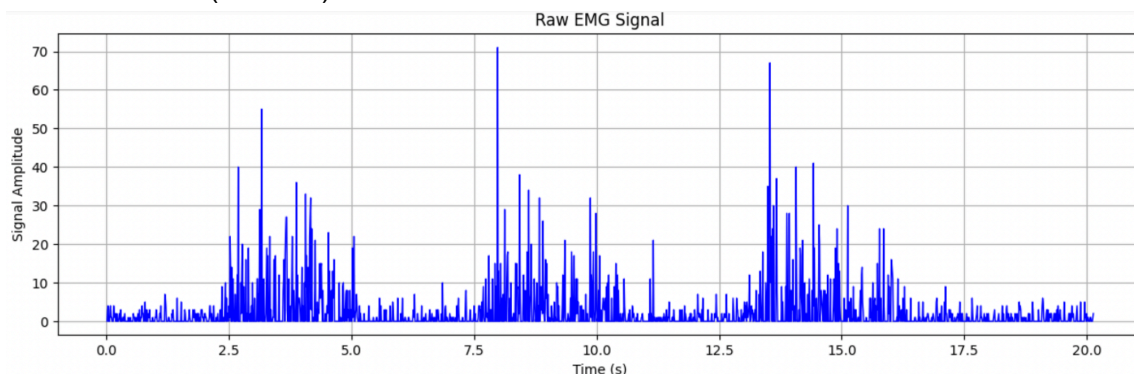
On Friday 1/31, we attempted to create a similar graph to this one from the **ABC of EMG** featuring a moving average filter, RMS filter, and butterworth filter:



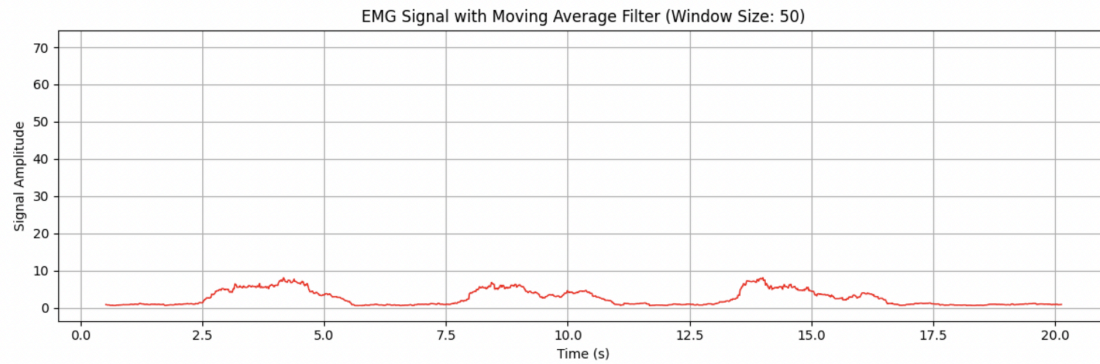
We collected raw EMG Signal data and wrote it to a file with CoolTerm. While collecting the data we realized our real-time Arduino graphs were probably plotting our data accurately, but the x-axis (the time) was just much visually wider than when plotted with Python, making it appear to have those more triangular spikes. We also noticed that our data contained a lot of zeros even during the moments when the muscle was flexed, but it looks like the reference data also did.

We then plotted the data with various filters via python. Moving average function was the `moving_average()` function, and RMS filter was `rms_filter()` function.

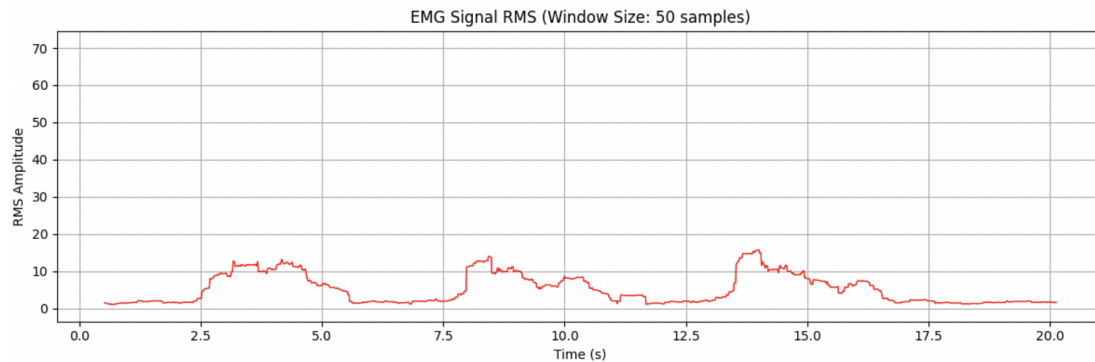
Raw EMG data (rectified):



Moving average filter:



RMS filter:



We also tried a butterworth filter with python's version of `filtfilt`, but it had many more adjustable parameters, and it ended up looking less promising than the other filters. (graph becomes extremely smooth at lower cutoff values).

```
# Butterworth filter
nyquist = fs / 2
normalized_cutoff = cutoff / nyquist
b, a = signal.butter(order, normalized_cutoff, btype='low')

# Apply forward-backward filter
filtered_signal = signal.filtfilt(b, a, raw_signal)
```

```
cutoff=400, fs=9600, order=2
```

